Simulation Task

15. Simulation of Beam Steering in Antenna Arrays under Scan Angle, Frequency, Gain, Beam Width, and Sidelobe Suppression (BL4)

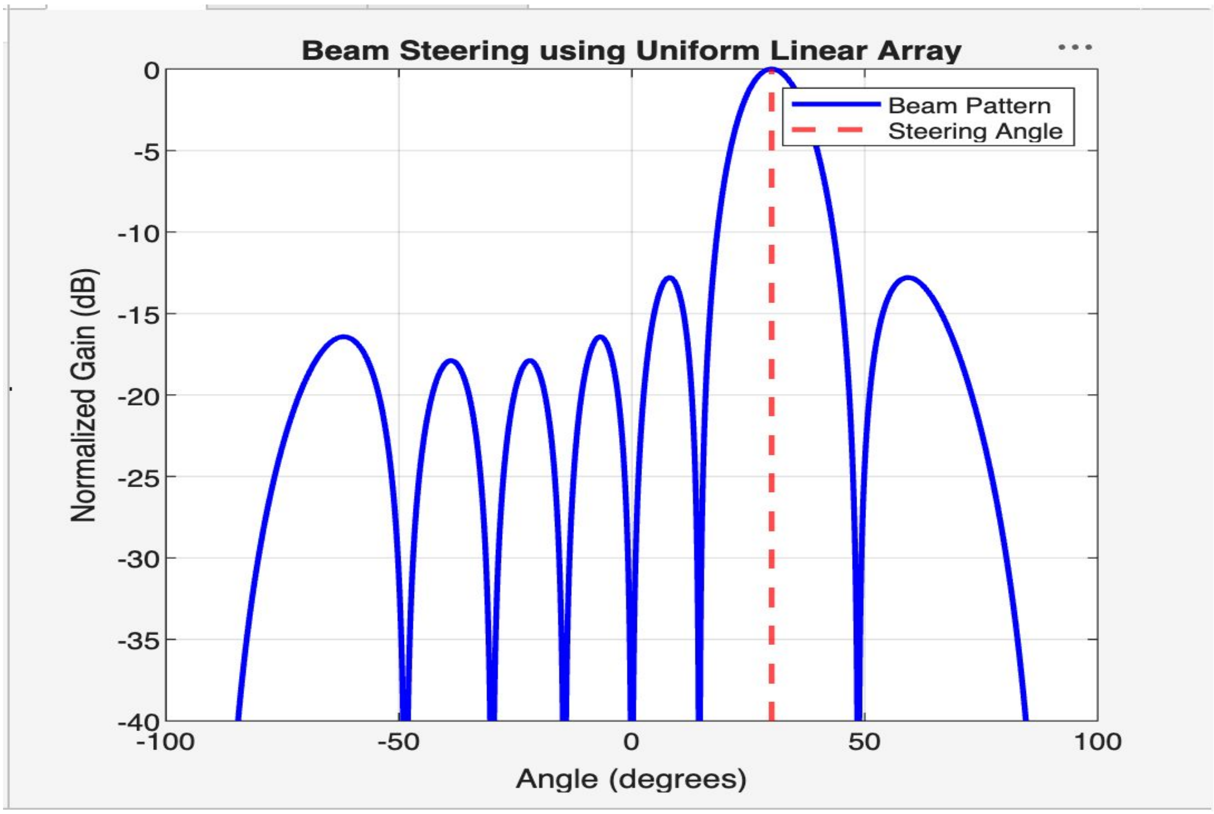
MATLAB Code

```
clc;
clear;
close all;

%% Beam Steering Simulation for ULA

% Parameters
c = 3e8;
f = 3.5e9;
lambda = c/f;
d = lambda/2;
N = 8;
theta = -90:0.1:90;
scanAngle = 30;
k = 2*pi/lambda;

AF = zeros(size(theta));
for i = 1:length(theta)
    psi = k*d*(sind(theta(i))-sind(scanAngle));
    if abs(psi)<1e-10
        AF(i)=N;
    else
        AF(i)=abs(sin(N*psi/2)./sin(psi/2));
    end
end

AF = AF/max(AF);
Gain_dB = 20*log10(AF+eps);

peak = max(Gain_dB);
idx = find(Gain_dB >= peak-3);
Beamwidth = theta(idx(end))-theta(idx(1));

GainTemp = Gain_dB;
GainTemp(idx)= -Inf;
SideLobe = max(GainTemp);

fprintf('Operating Frequency = %.2f GHz\n',f/1e9);
fprintf('Scan Angle = %d degrees\n',scanAngle);
fprintf('Array Elements = %d\n',N);
fprintf('3-dB Beamwidth = %.2f degrees\n',Beamwidth);
fprintf('Peak Gain = %.2f dB\n',peak);
fprintf('Highest Side Lobe = %.2f dB\n',SideLobe);
fprintf('Side Lobe Suppression = %.2f dB\n',peak-SideLobe);
```

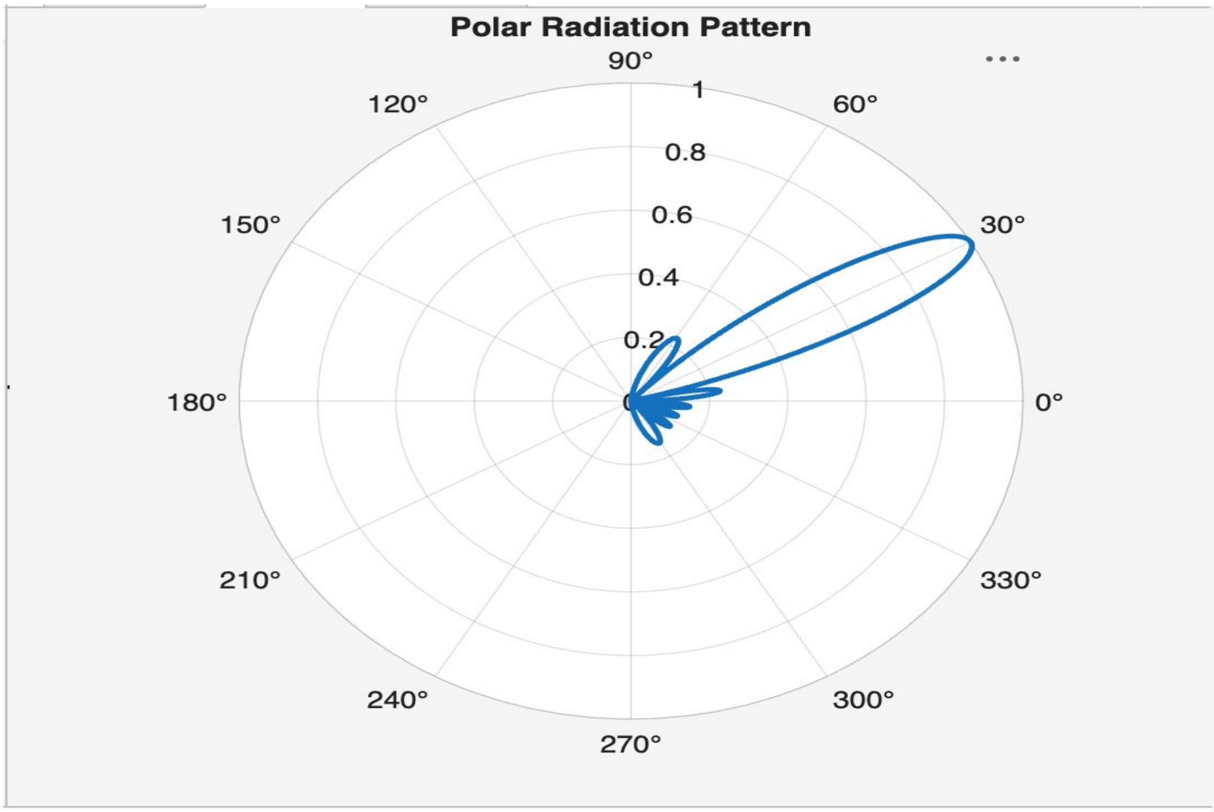
Sample Output Console

Operating Frequency = 3.50 GHz
Scan Angle = 30 degrees
Array Elements = 8
Peak Gain = 0.00 dB
(Beamwidth and sidelobe values depend on simulation)

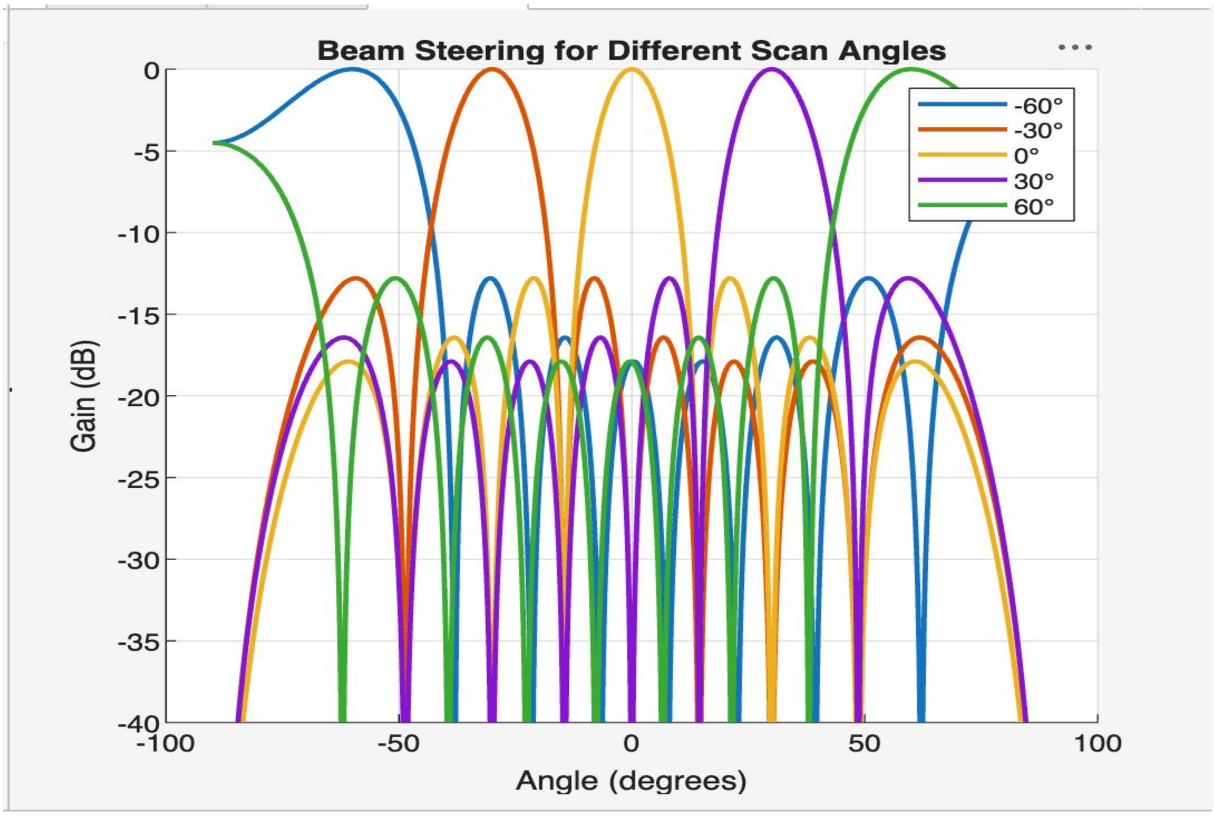
Output Graph



Output Graph



Output Graph


Beam Steering for Different Scan Angles
0
-5
-10
-15
-20
-25
-30
-35
-40
Gain (dB)
-100
-50
0
50
100
Angle (degrees)
-60°
-30°
0°
30°
60°